

Solve the equation

$$x+5 = 14 - \frac{1}{2}x$$

Notes

1) Solve the equation $\approx$ find x such that satisfies the relationship $x+5 = 14 - \frac{1}{2}x$

2) $x+5 = 14 - \frac{1}{2}x$ [original equation]

$$\Rightarrow x+5 + \frac{1}{2}x = 14 - \frac{1}{2}x + \frac{1}{2}x \quad [\because ?? \text{ property}]$$

$$\Rightarrow x+5 + \frac{1}{2}x = 14 + (-\frac{1}{2}x + \frac{1}{2}x) \quad [\because \text{associativity}]$$

$$\Rightarrow x+5 + \frac{1}{2}x = 14 + 0 = 14$$

$$\Rightarrow x + \frac{1}{2}x + 5 = 14 \quad (\because \text{transitivity})$$

$$\Rightarrow \frac{3}{2}x + (5-5) = 14-5 = 9 \quad (\because \text{associativity})$$

$$\Rightarrow \frac{3}{2}x + 0 = 9$$

$$\Rightarrow \frac{3}{2}x = 9$$

$$\Rightarrow 2 \cdot \frac{3}{2}x = 2 \cdot 9 = 18$$

$$\Rightarrow 3x = 18 \quad (\because \text{associativity})$$

$$\Rightarrow \left(\frac{1}{3} \cdot 3\right) \cdot x = \frac{1}{3} \cdot 18 = 6$$

$$\Rightarrow \boxed{x = 6}$$